

RMD-Q: Restricted Module Decoding with Quadratic Constraint

— A New Post-Quantum Assumption with Lightweight KEM and Signature Constructions, Analysis, and Verified Prototype

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(DRAFT v0.1 — research report, not a standard)

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Abstract

We introduce *Restricted Module Decoding with Quadratic Constraint* (RMD-Q), a new post-quantum hardness assumption that jointly couples a noisy module-linear relation $b = As + e$ over $R_q = \mathbb{Z}_q[X]/(X^n + 1)$ with a sparse multivariate-quadratic constraint $P(s) = 0$ over the same secret: breaking an instance requires a lattice decoder *and* an MQ solver together, neither alone suffices. From one assumption we build both a Fujisaki-Okamoto key-encapsulation mechanism and a Fiat-Shamir signature with dual lattice/MQ openings, sharing ring arithmetic, XOF layer, and codebase (crypto core ≈ 3.9 KB). This report documents the assumption (including a plantable key-generation distribution with measured uniformity), exact decrypt-failure distributions by convolution (down to 2^{-910} uncompressed, 2^{-235} at Kyber-style compression), dual-BKZ and hybrid cost models validated within 3 bits on ML-KEM-512, a calibrated kill-phase campaign (toy instances fall to brute force, MITM, LLL, Gröbner, and msolve; larger instances stall or exhaust a 16-core/31 GB box), and a constant-time-shaped C implementation cross-checked byte-for-byte against an independent Python reference (NTT backend with exhaustively proven Barrett reduction, sanitizers and valgrind clean). We make *no* NIST category claims and *no* deployment recommendation: the assumption is conjectured, several analyses are explicitly open, and Section 7 lists everything this report does *not* claim. All artifacts needed to reproduce every number are described in Section 7.

Keywords: post-quantum cryptography, module-LWE, multivariate cryptography, planted problems, key encapsulation, digital signatures, cryptanalysis, verified implementation

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1 Introduction

1.1 Motivation

The NIST post-quantum standards (ML-KEM [1], ML-DSA [2], SLH-DSA [3]) secure the general case, but leave two gaps this work addresses as a *research contribution, not a deployment proposal*: (i) ultra-constrained targets where every kilobyte of code and every millijoule counts, and (ii) assumption diversity beyond pure Module-LWE / Module-SIS and hash-based designs (cf. the ongoing additional-signatures process [5]).

1.2 Contributions

1. **A new assumption (Section 3).** Search/Decision RMD-Q: sparse-short s with $b = As + e$ and $P(s) = 0$ for a sparse homogeneous-quadratic system P . We give a plantable key-generation distribution (always terminates, unlike rejection sampling at scale) with measured coefficient uniformity ($\chi^2 = 17.4$ vs. null mean 15) and a documented leakage analysis.
2. **Two constructions from one assumption (Sections 4, 4.2).** An FO-KEM (full encaps/decaps with implicit rejection running in C at $n = 256$) and a Fiat-Shamir signature with dual lattice/MQ openings (toy status, Section 4.2 states its limits). A found-by-testing soundness restriction (homogeneous-quadratics only) is documented with its root cause.
3. **Quantified analysis, no category claims (Section 5).** Exact failure distributions by convolution (2^{-910} uncompressed; 2^{-235} at $(d_u, d_v) = (10, 4)$); dual-BKZ model validated within 3 bits on ML-KEM-512 (20-bit optimism at 768, stated); guessing+lattice hybrid model (optimum: no guessing at these densities; wt lever quantified); joint XL model (MQ leg contributes negligibly in-model); Gröbner/msolve scaling curve with a 114 GB OOM data point; 200 000 C-speed failure trials, zero failures.
4. **Verified prototype (Section 6).** Independent C and Python implementations cross-checked byte-for-byte on all KATs; NTT backend with exhaustively proven Barrett reduction; constant-time twins with automated timing smoke-test; ASan/UBSan/valgrind clean; Cortex-M4 cross-compile clean.

1.3 What this report is not (see Section 7)

Not a standard, not a submission, not deployment guidance, not a proof. Security categories are never claimed; several analyses are explicitly open; Section 7 maps every claim to evidence and blockers.

2 Preliminaries and notation

Let $R_q = \mathbb{Z}_q[X]/(X^n + 1)$ with n a power of two and q prime; our production ring is $(n, q) = (256, 3329)$, the standard lattice ring shared with ML-KEM [1] (deliberately: novelty is claimed only for the joint assumption and constructions, never the arithmetic). $S(\eta, w)$: each polynomial has at most w nonzero coefficients, values uniform over $[-\eta, \eta]$ then sparsified (reference `sample_sparse_poly`); secrets only. $\text{CBD}(\eta)$: Kyber-style centered binomial, fixed-shape sampling; errors and encryption randomness. $U(\mathbb{Z}_q)$: uniform; matrix A expansion. Symmetric layer is FIPS-202 SHAKE128/256 only [4], with fixed domain separation (matrix expand, FO coins/shared-secret, signature hashes, implicit reject). KEM security target: IND-CCA2 via the modular Fujisaki-Okamoto transform [6]. Signatures follow Fiat-Shamir with aborts [7, 8]. BKZ cost models used: $2^{0.292\beta}$ classical (BDGL sieve [9]), $2^{0.265\beta}$ quantum, $2^{0.2075\beta}$ paranoid.

3 The RMD-Q assumption

Definition 1 (Search-RMD-Q). *Parameters $(n, q, k, \eta, w, \eta_e, t, m)$, $N = k \cdot n$. Gen samples $A \leftarrow U(R_q^{k \times k})$ (SHAKE128), $s \leftarrow S(\eta, w)^k$, $e \leftarrow \text{noise}(\eta_e)$, $P \leftarrow \text{MQ}(t, m)$ conditioned on $P(s) = 0$, and sets $b = As + e$. Given $x = (A, b, P)$, find s' with $\text{wt}(s') \leq k \cdot w$, $\|s'\|_\infty \leq \eta$, $\|As' - b\|_\infty \leq B_e$, and $P(s') = 0$ exactly. Decision-RMD-Q distinguishes $(A, As + e, P)$ from uniform.*

The assumption is that breaking an instance requires a lattice decoder *and* an MQ solver *jointly*: lattice-only candidates fail $P(s') = 0$ with probability $\approx 1 - q^{-t}$ each, while MQ-only roots violate the noisy linear bound. No Shor structure exists (random module matrix, random sparse MQ); Grover gives at most a square-root speedup over the $C(N, w) \cdot (2\eta + 1)^w$ sparse space.

3.1 Plantable key generation

Rejection-sampling s until $P(s) = 0$ succeeds with probability $\approx q^{-t}$ per try – impossible at scale. Our planted distribution samples s first, then per equation samples $m-1$ free terms (uniform supports and coefficients over $[0, q)$) plus one fix monomial on $\text{support}(s)$ with solved coefficient $c_{\text{fix}} = -\text{rest} \cdot \text{ev}^{-1} \bmod q$, inserted at a uniform slot; free coefficients are resampled (bounded) until $c_{\text{fix}} \neq 0$, with a pure-random escape hatch when all free monomials vanish at s (then uniform-random, no conditioning at all). Given supports and s , coefficients are uniform over the hyperplane $\{c : \langle c, \text{evals} \rangle = 0\}$ minus a $1/q$ slice. Measured: fix-coefficient $\chi^2 = 17.4$ on 209 samples over $\mathbb{Z}_{17}^*$ (null mean 15); positions spread over $\text{supp}(s)$. Accepted residual leakage (pad linear terms on zero coordinates; $1/q$ conditioning mass) is documented, not claimed away. First-ever joint keygen at $(256, t=8)$ runs instantly with passing KEM round-trips. This shifts random-MQ to *planted*-MQ – same family as planted instances in MPCitH/code-based designs, but our sparse-planted distribution needs dedicated analysis (Section 5).

4 Constructions

4.1 KEM (FO-KEM over RMD-Q CPA-PKE)

The shape mirrors FIPS 203 (CPA-PKE + Fujisaki-Okamoto $\Rightarrow$ IND-CCA2 [6]) with the RMD-Q relation as trapdoor; P participates in keygen and domain separation (ciphertexts carry no MQ opening in v0.1 – a documented limitation, not a security argument). **KeyGen** expands A , samples sparse s , plants P , samples CBD error e , sets $b = As + e$. **Encaps** derives coins as $\text{SHAKE256}(\text{coins} \parallel m \parallel \text{pkhash})$, samples dense CBD (r, e_1, e_2) , and outputs $u = A^T r + e_1$, $v = b^T r + e_2 + \text{encode}(m)$ with K derived alongside. **Decaps** decodes $m' = v - s^T u$, re-encrypts, byte-compares, and returns K or $\text{SHAKE256}(\text{reject} \parallel \sigma \parallel ct)$ (implicit rejection). Correctness needs per-coefficient noise $< q/4$; exact failure distributions are computed in Section 5.

4.2 Signature v1 (toy status)

Same keypair shape; Fiat-Shamir with a *dual* opening from one mask y : $z_{\text{lat}} = y + c_{\text{poly}} s$ (sparse weight- τ challenge) for the lattice check $w' = Az_{\text{lat}} - c_{\text{poly}} b \approx w$, and $z_{\text{mq}} = y + c_{\text{scalar}} s$ with bilinear opening h satisfying $P(z_{\text{mq}}) = P(y) + ch$ for the MQ check. **Load-bearing restriction (found by testing)**: P must be homogeneous-quadratic – with linear parts the opening gains a $c^2 P_{\text{quad}}(s)$ defect and verification fails in general; toy sparsity masked this (both parts vanished independently by luck). The opening asserts $v_j \neq -1$. Unspecified in v0.1: prod challenge (τ, γ_1, β) with size analysis, wire sizes, hedged policy, rejection-rate analysis, $n=256$ C code, and any zero-knowledge property.

4.3 Parameters

set	n	q	k	η	w/poly	η_e	t	m	wire
TOY-16	16	17	1	1	4	1	2	4	raw
TOY-32	32	97	1	1	6	1	3	6	raw
PROD-256	256	3329	2	2	128	2	8	12	12-bit
PROD-384	256	3329	3	2	160	2	10	12	12-bit (no KATs)

The ring is shared with ML-KEM deliberately (NTT reuse, mature ring cryptanalysis, comparability); $k=2/3$ matches ML-KEM-512/768 dimensions; $w=128$ closes the modeled 5-bit gap to the Kyber reference at $\approx$ zero size cost (Section 5); t/m supported by Gröbner scaling and XL-infeasibility results below. Wire v0.1 is uncompressed 12-bit packing (pk $\approx$ 0.8 KB, ct $\approx$ 0.8–1.1 KB); compressed (10, 5) FO path (800 B ct, re-compress-then-compare decapsulation) proven end-to-end in Python with a byte-exact C codec.

5 Security analysis: models, measurements, kill-phase

All numbers below are directional (model tolerances stated); each result points at its reproduction artifact (Section 6).

Lattice leg. Dual-BKZ model (Bai-Galbraith scaled dual; $2^{0.292\beta}/2^{0.265\beta}/2^{0.2075\beta}$ costs) validates within 3 bits on ML-KEM-512 ($2^{115}/2^{104}$ vs. published $2^{118}/2^{107}$; 20-bit optimism at 768, stated). Ours: $k=2 \approx 2^{110}/2^{99}$, $\approx 5\text{--}7$ bits under same- N reference in-model. fpylll cross-check of the BKZ constant agrees within 0.3% (conservative direction).

Hybrid guessing. Position+value guessing + BKZ remainder: optimum is $g=0$ at all densities (guessing uneconomical at 25% density); enumeration validated empirically; weight lever quantified ($w=256$ total closes the gap). Perfect-hints curve: ≈ 10 bits lost per 32 leaked coordinates. Unhandled: hints-variants, BKW, quantum walks.

Joint lattice+MQ. XL with field equations counted attacker-favorably is infeasible at prod scale (never closes at any degree ≤ 14); lattice path binds in-model. Planted-structure-specific attacks remain unmodeled – the open cryptanalytic question.

Gröbner/msolve kill-phase. Planted $n=16$: basis 0.3–2s; full *solve* >300 s even at $n=16$ (bases cheap, solutions hard). $n=24$ basis 180 s, $n=28$ basis >1200 s (bounded, unfinished), $n=32$ basis exhausts 114 GB on 16 cores (kernel-logged OOM – memory wall, not timeout). Pure-MQ (no bounds) terminates trivially everywhere but recovers nothing (positive-dimensional variety). Planted vs. random show no measurable non-genericity at $n=16$; Bardet bounds uninformative for this shape (predicted 15/27 vs. observed 2–4). Toy instances additionally fall to brute force, MITM (exact key, 23M checks in C), and LLL+Babai hybrids.

Failures. Exact convolution distributions (FIPS-203 style): $2^{-910}\text{--}2^{-512}$ per-KEM uncompressed across weights; $2^{-268.6}$ at $(d_u, d_v) = (10, 5)$, $2^{-235.4}$ at $(10, 4)$ – far under the 2^{-138} bar. 200 000 C-speed trials at $n=256$ (both paths): zero failures (Clopper-Pearson 95% upper $\approx 2^{-15}$).

6 Implementation and verification

Independent C and Python implementations cross-checked byte-for-byte on all KATs (matrix expansion, sampling, ciphertexts, shared secrets, implicit-reject keys, signature vectors). NTT backend for (256, 3329) (verified schedule: forward pairs equal direct CRT reduction; full multiply equals schoolbook): $6.0\times$ multiply, $8.8\times$ matrix-vector ($\approx 25\mu\text{s}$, full KEM $\approx 0.25\text{ ms}$ host; $\approx 1.6\text{M}$ M4 cycles projected). Barrett reduction exhaustively proven over all 11M products. Constant-time twins (masked arithmetic, Barrett-only, inverse table) equivalence-tested with an automated objdump smoke check (zero sign-jumps; reference still shows its jump, proving sensitivity); CT costs $\approx 1.05\text{--}1.17\times$. CBD sampling wired into FO paths; sparse legacy retired behind `-DRMDQ_NO_LEGACY` (hardened-only build fully green). ASan/UBSan/valgrind clean; strict warnings plus analyzer silent; Cortex-M4 `-Werror` cross-compile clean (crypto core $\approx 3.9\text{ KB}$; NTT scratch 24KB shared – too fat for small MCUs, streaming path retained). Sizes: pk $\approx 0.8\text{ KB}$, ct $\approx 0.8\text{ KB}$ uncompressed. One-command reproduction: `run_all.sh`.

7 Non-claims, gaps, and roadmap

Not claimed: any NIST category; any reductionist proof (no ROM/QROM sketches exist); certification of RMD-Q/planted-MQ; side-channel resistance beyond measured timing smoke; any signature security level; deployment readiness (no audit, single-threaded statics, variable-time Python reference). Model numbers are directional with stated tolerances.

Blocking gaps: joint Gröbner degree-of-regularity at scale; proper noisy-hint framework; BKW/quantum-walk analyses; fpvll full-attack cross-check; exact failure confirmation via C trials at scale; compressed wire KATs; signature prod parameters and proofs; formal object-level timing proof; external review.

8 Conclusion

RMD-Q couples lattice and multivariate hardness in one assumption supporting both KEM and signatures from a shared lightweight codebase, with measurement-backed analysis and an explicit, checkable list of what remains unknown. We invite attack on every open item – that scrutiny is the point of this report.

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